IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [The Daily Air Quality Action Agent]

Domain of Project – [aiairguard.com]

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Problem Statement -

Problem Statement No.15: The Daily Air Quality Action Agent

The Challenge

Air pollution is a **major public health crisis**, contributing to respiratory issues, climate change, and reduced quality of life. Citizens often receive air quality index (AQI) data, but it is presented in **technical formats**, making it difficult to understand or translate into **daily lifestyle actions**. Traditional monitoring systems lack **personalized recommendations** and proactive alerts.

The Objective

Build an **AI-powered Daily Air Quality Action Agent** that translates **air quality data into actionable insights** for citizens, policymakers, and organizations. The solution should:

1. **AQI Data Retrieval (RAG) –** Collect and interpret real-time AQI from public APIs, satellite data, or government sources.
2. **Personalized Health Recommendations** – Provide daily actionable advice (e.g., avoid outdoor exercise, use masks, keep children indoors) tailored to age, location, and health conditions.
3. **Predictive Alerts –** Forecast poor air quality days, issue early warnings, and suggest preventive measures like adjusting commute or using air purifiers.
4. **Policy & Awareness Insights –** Suggest community-level actions (e.g., carpooling, waste reduction) and summarize policy advisories for public awareness.
5. **Multi-Agent System** -
   - AQI Data Agent (fetches & interprets data)
   - Health Advisory Agent (personalized lifestyle suggestions)
   - Forecasting Agent (predicts trends & risks)
   - Community Action Agent (drives awareness campaigns)
6. **Visualization Dashboard** – Show daily AQI levels, risk alerts, and recommended actions in a clear and interactive format.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Proposed Solution -

Proposed Solution: AI AirGuard — Daily Air Quality Action Agent

AI AirGuard is an intelligent, web-based air quality monitoring system that helps people understand pollution levels and make informed decisions to protect their health. The system retrieves air quality data for selected cities and displays the Air Quality Index (AQI), PM2.5, PM10, and other available pollutant measurements through an interactive dashboard.

Based on the current AQI, the system provides personalized health recommendations, pollution alerts, and practical actions to reduce exposure to unhealthy air. It also includes visual charts to understand pollution trends and an AI-powered assistant to answer questions about air quality and pollution prevention.

The proposed solution combines real-time data integration, intelligent recommendations, and an easy-to-use interface to make air quality information more accessible and actionable. If live data or AI services are unavailable, a clearly labelled demo mode allows users to explore the system's features.

Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually

- **IBM Grainte model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization

- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.

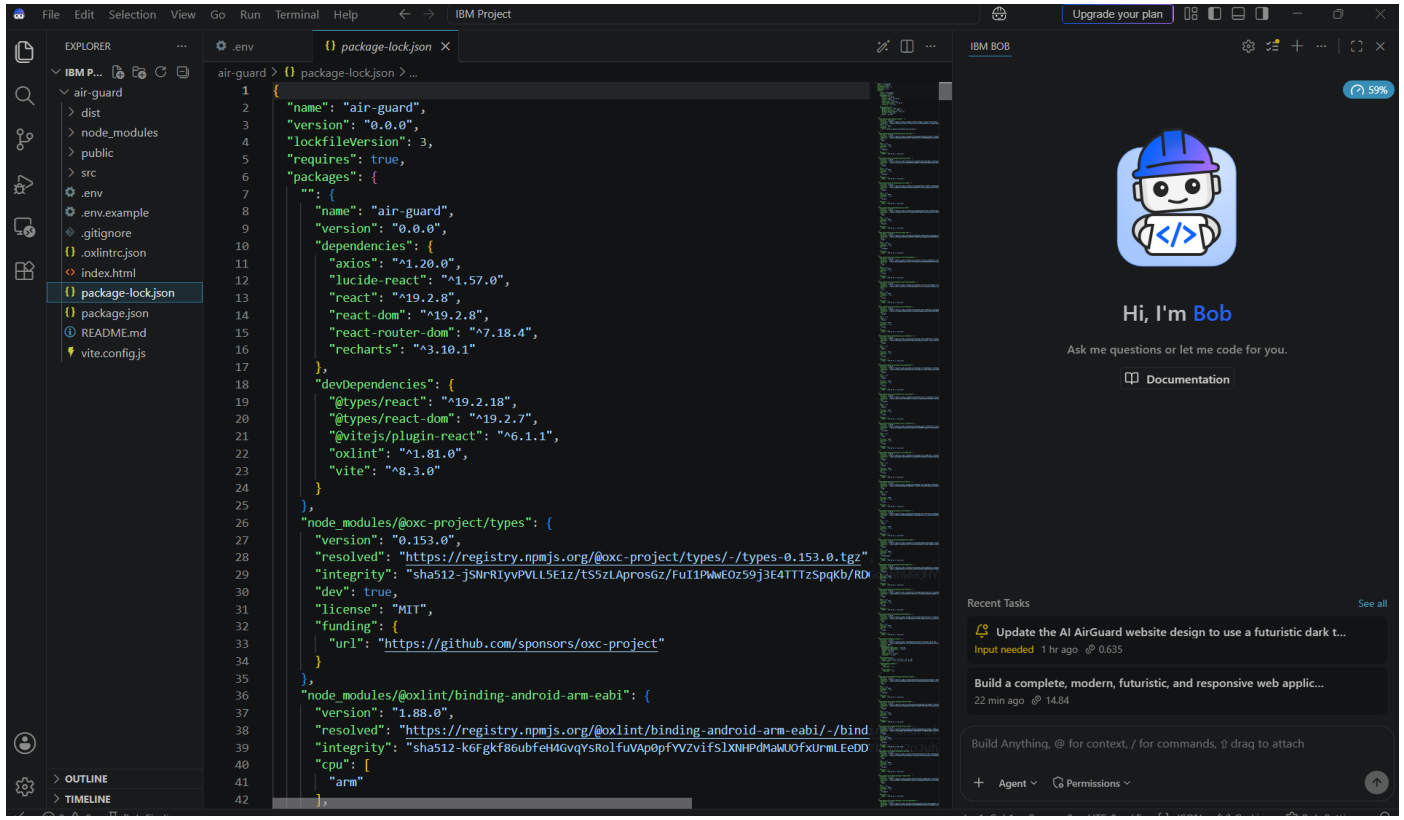
- **RAG–** RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.

- **IBM Watsonx.ai -** For model deployment, embeddings, and AI governance

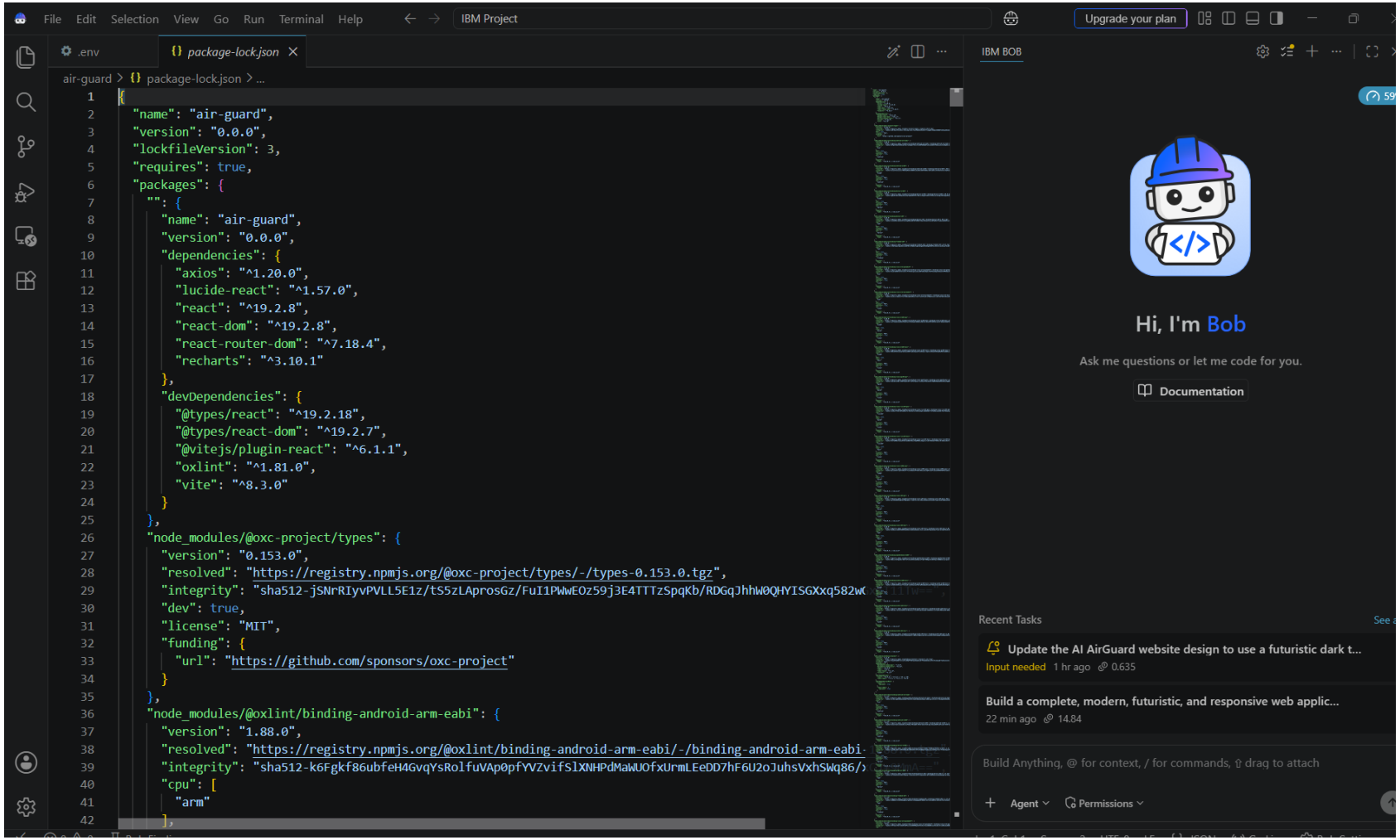
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

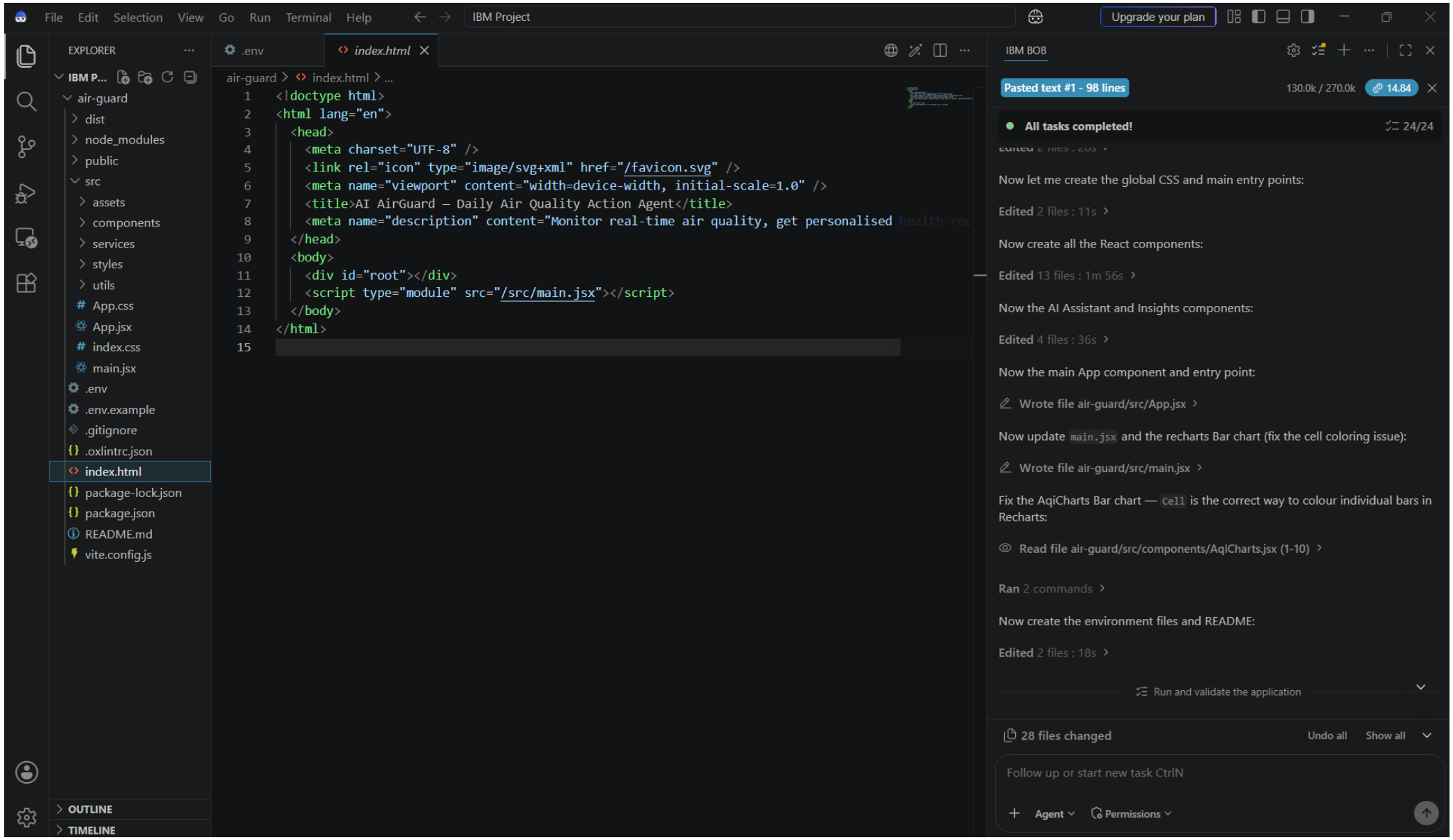
Required Files created using IBM bob -

1. Air-guard
2. .env.example
3. .gitignore
4. .oxlintrc.json
5. Index.html
6. Package-json
7. REASME.md
8. Vit.config.js

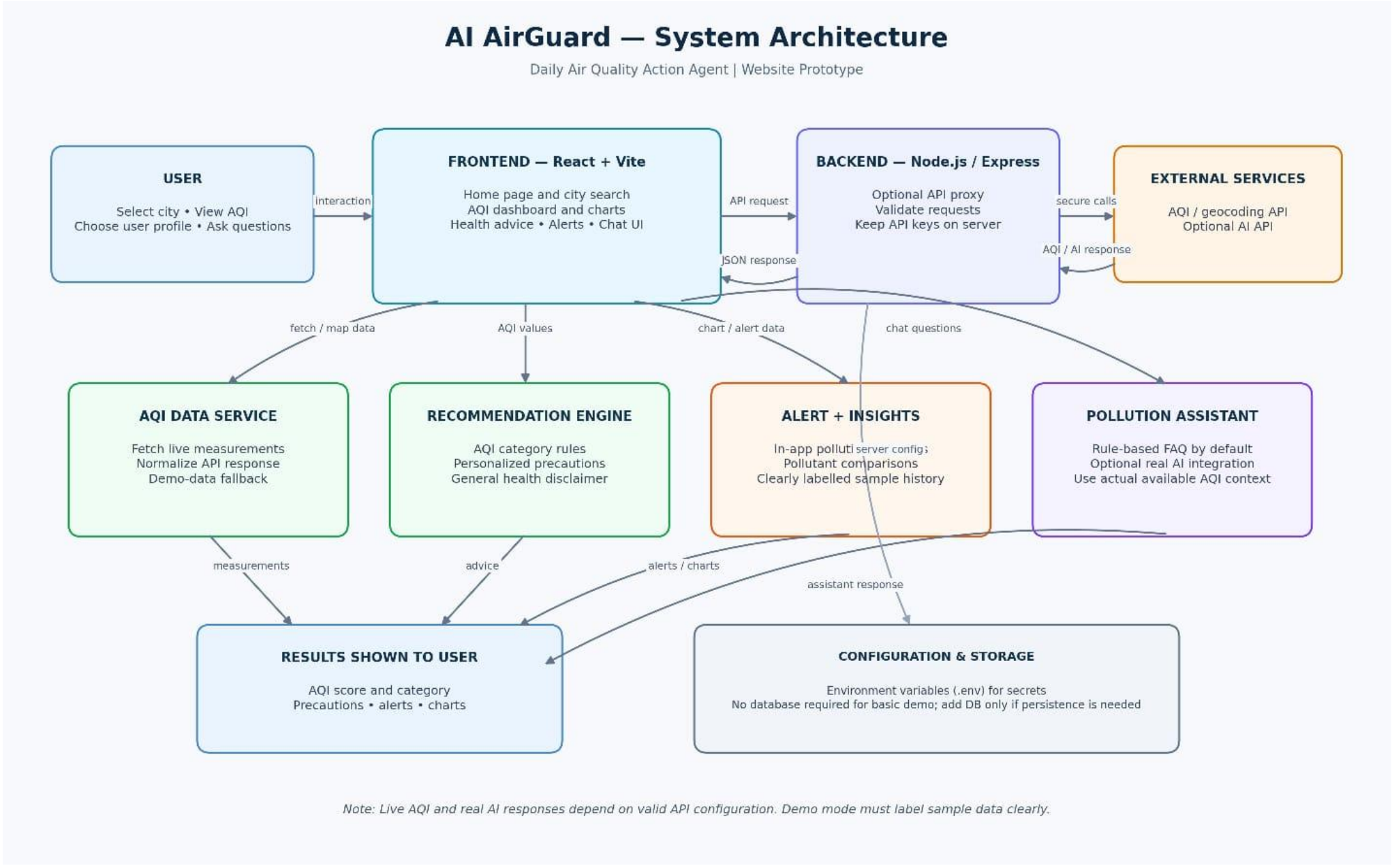


IBM Bob Screenshots:




File Edit Selection View Go Run Terminal Help
IBM Project
Upgrade your plan
EXPLORER
IBM P...
air-guard
dist
node_modules
public
src
assets
components
services
styles
utils
App.css
App.jsx
index.css
main.jsx
.env
.env.example
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
README.md
vite.config.js
OUTLINE
TIMELINE
.env
index.html
air-guard > index.html > ...
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
6 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7 <title>AI AirGuard – Daily Air Quality Action Agent</title>
8 <meta name="description" content="Monitor real-time air quality, get personalised
9 </head>
10 <body>
11
12 <script type="module" src="/src/main.jsx"></script>
13 </body>
14 </html>
15
IBM BOB
Pasted text #1 - 98 lines
130.0k / 270.0k
14.84
All tasks completed!
24/24
Now let me create the global CSS and main entry points:
Edited 2 files : 11s
Now create all the React components:
Edited 13 files : 1m 56s
Now the AI Assistant and Insights components:
Edited 4 files : 36s
Now the main App component and entry point:
Wrote file air-guard/src/App.jsx
Now update main.jsx and the recharts Bar chart (fix the cell coloring issue):
Wrote file air-guard/src/main.jsx
Fix the AqiCharts Bar chart — Cell is the correct way to colour individual bars in Recharts:
Read file air-guard/src/components/AqiCharts.jsx (1-10)
Ran 2 commands
Now create the environment files and README:
Edited 2 files : 18s
Run and validate the application
28 files changed
Undo all
Show all
Follow up or start new task CtrlN
Agent
Permissions

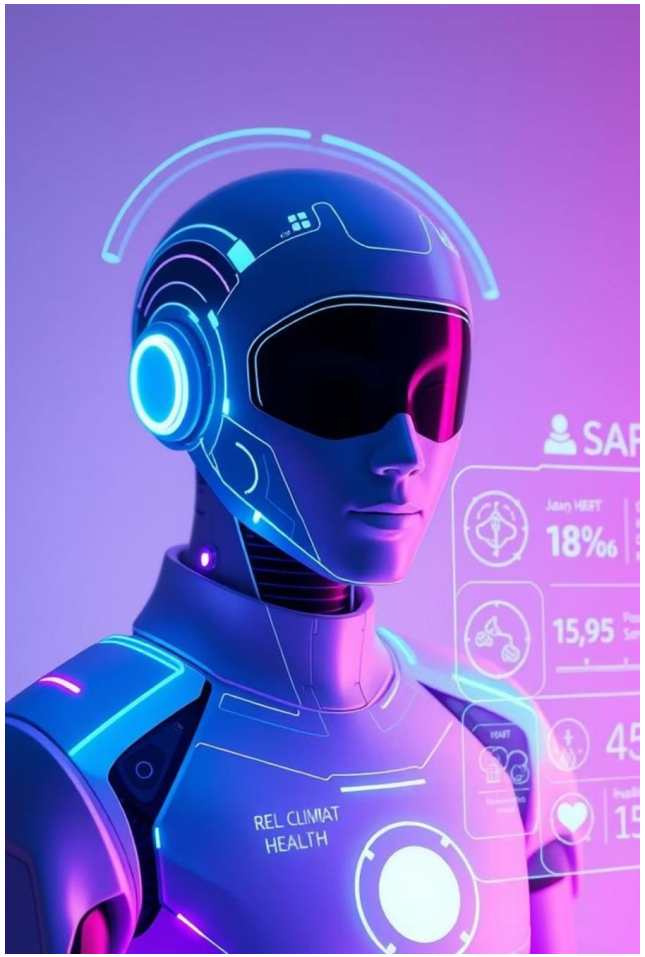
Architecture blueprint for IBM bob:



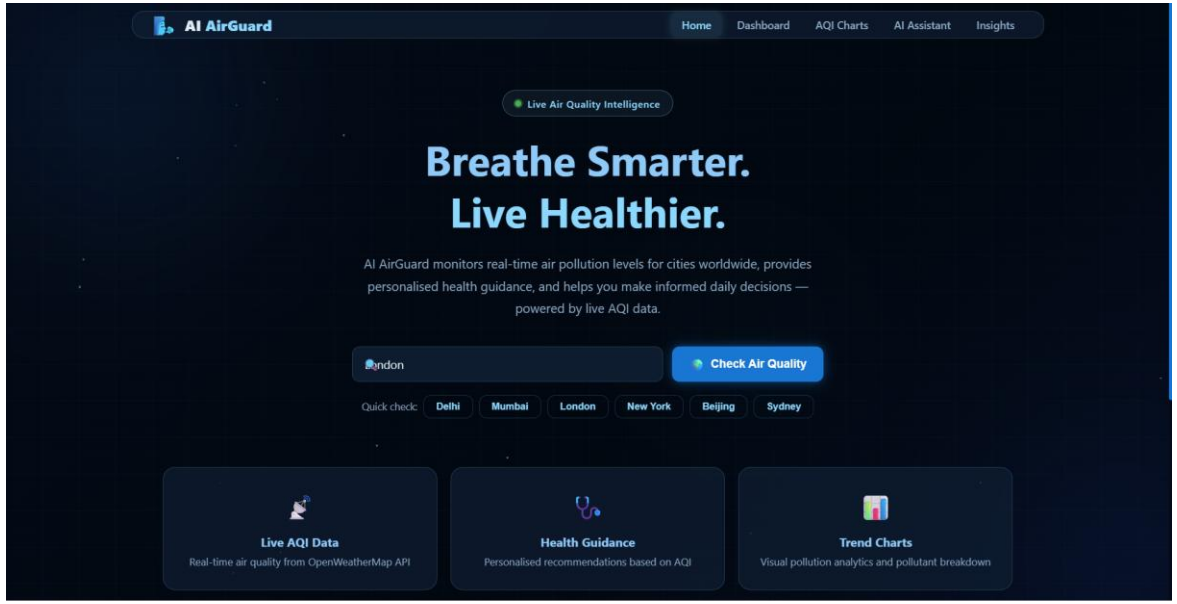
Role of Agentic AI in the solution

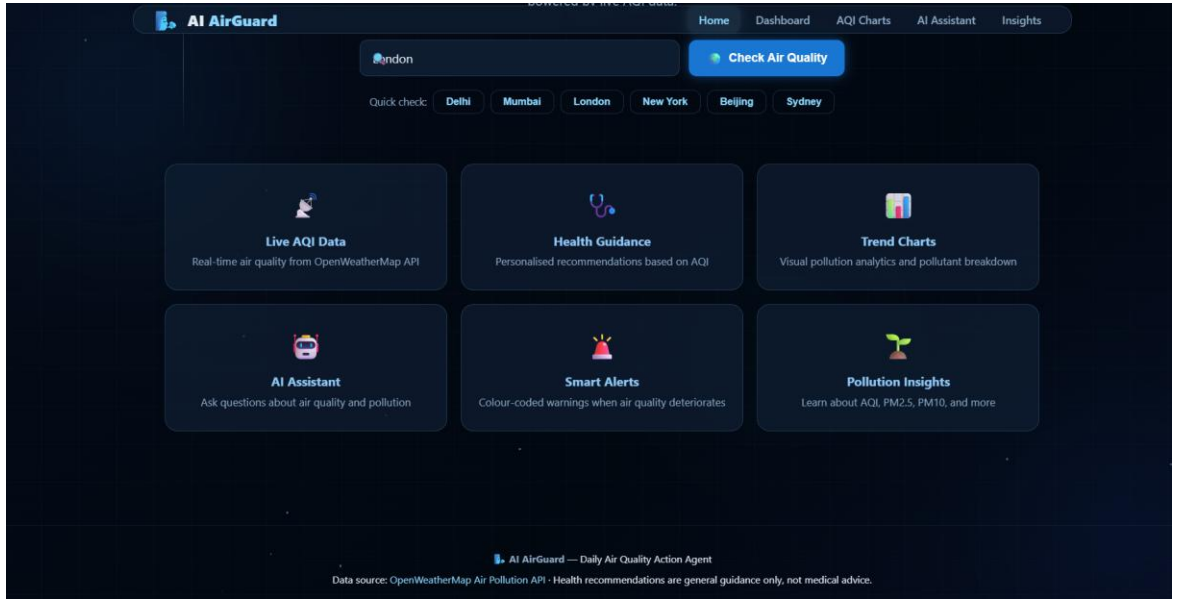
Role of Agentic AI in AI AirGuard

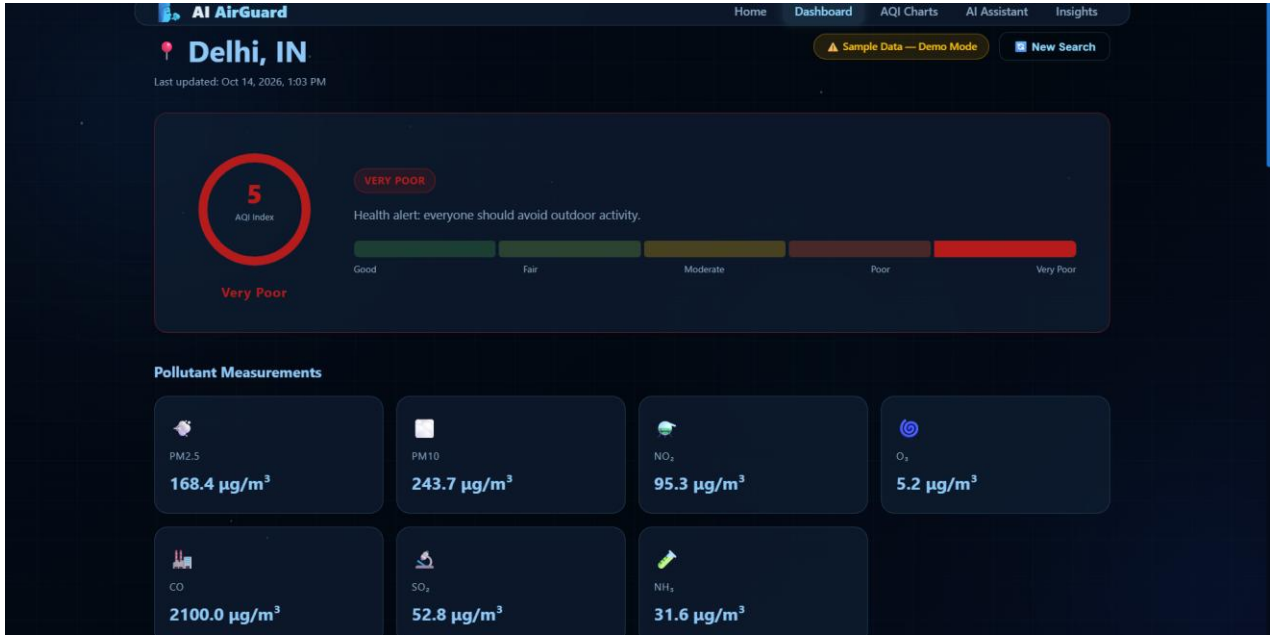
AI AirGuard uses Agentic AI to monitor air quality data, analyze pollution levels, and recommend appropriate actions. The agent retrieves AQI information for the selected city, identifies pollution severity, and generates personalized health precautions based on the user's profile. It can trigger alerts when pollution levels become unhealthy and answer user questions through an intelligent pollution assistant. By combining real-time data, decision-making rules, and automated recommendations, the system helps users make informed decisions and reduce their exposure to air pollution.



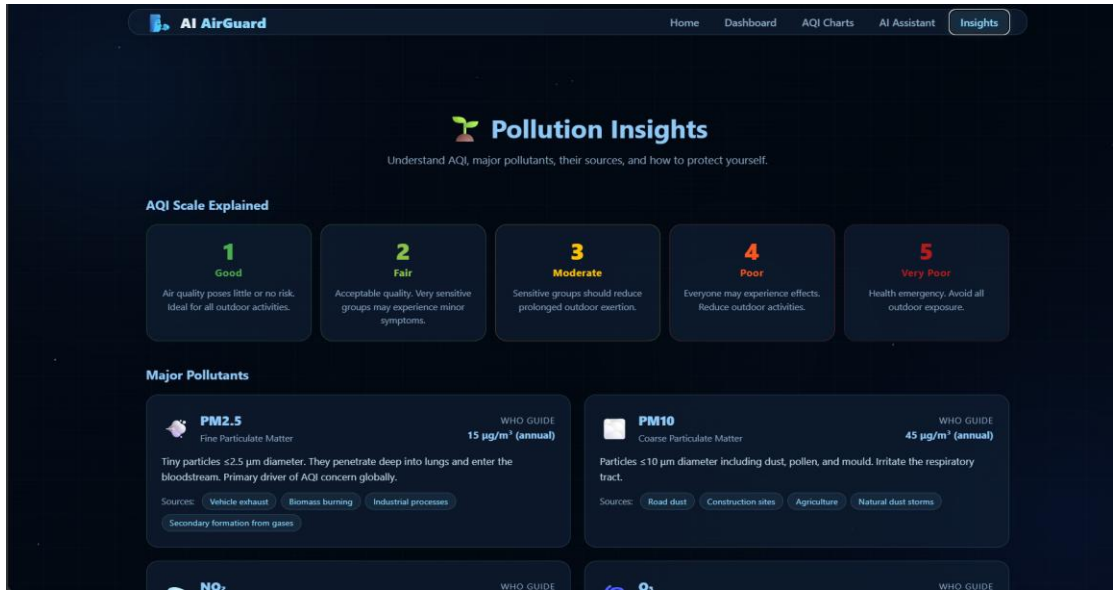
Project Output Screenshot

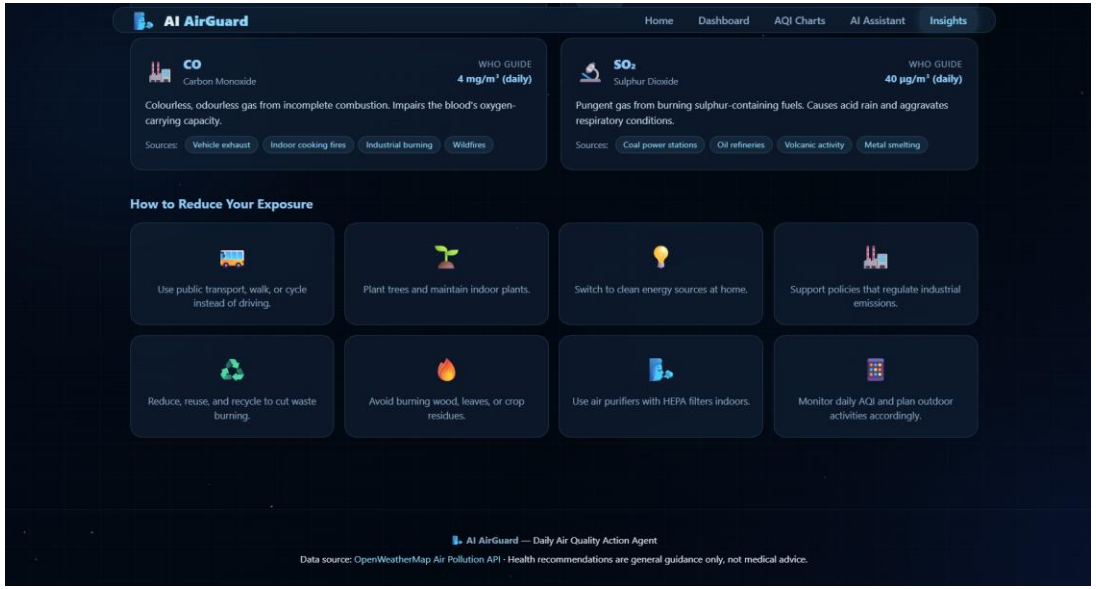


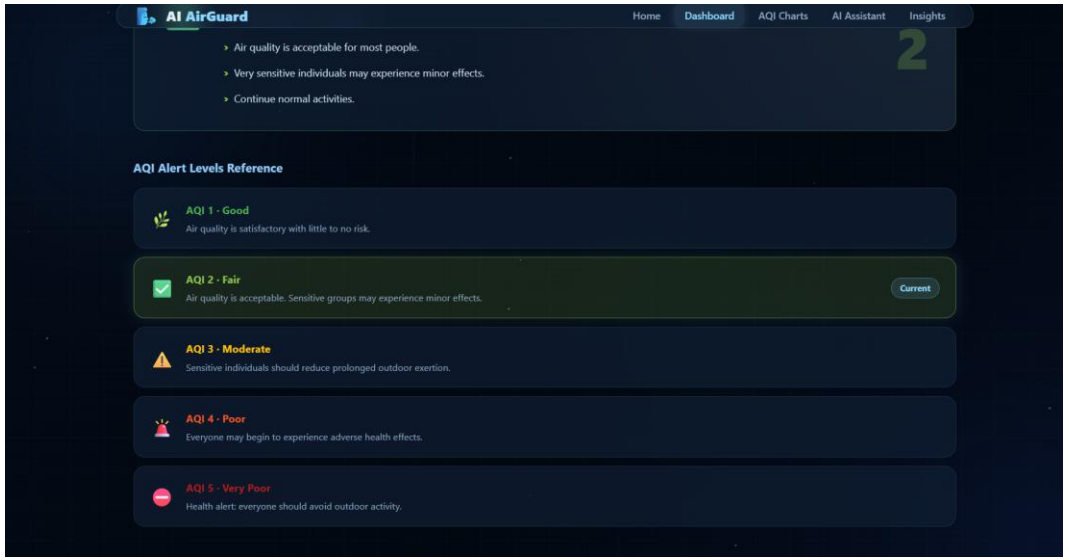




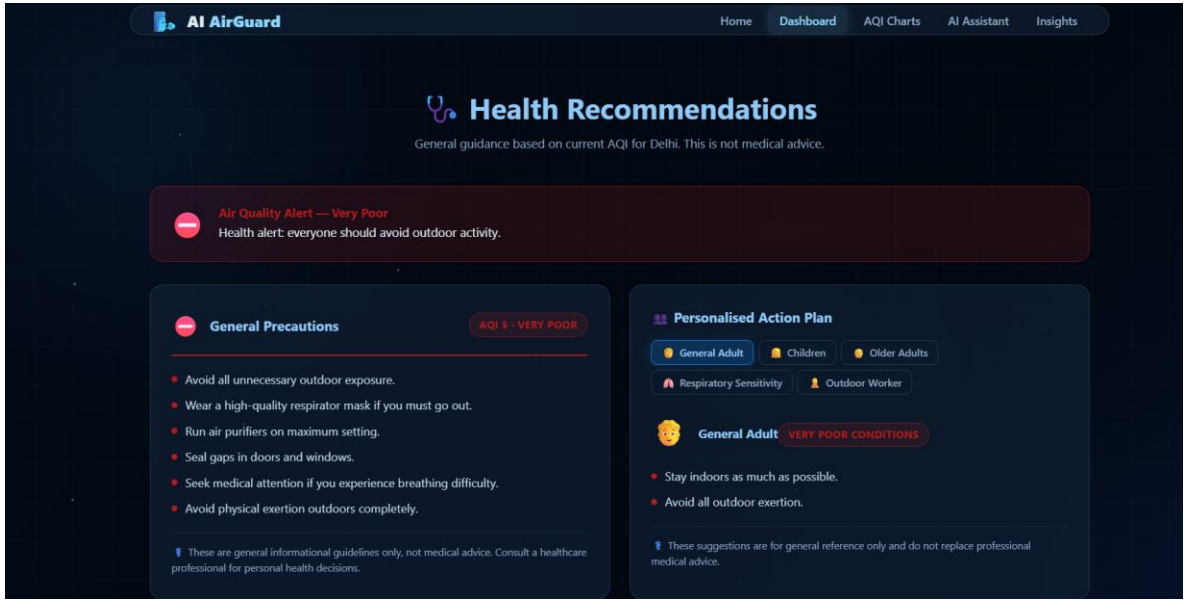
Project Output Screenshot



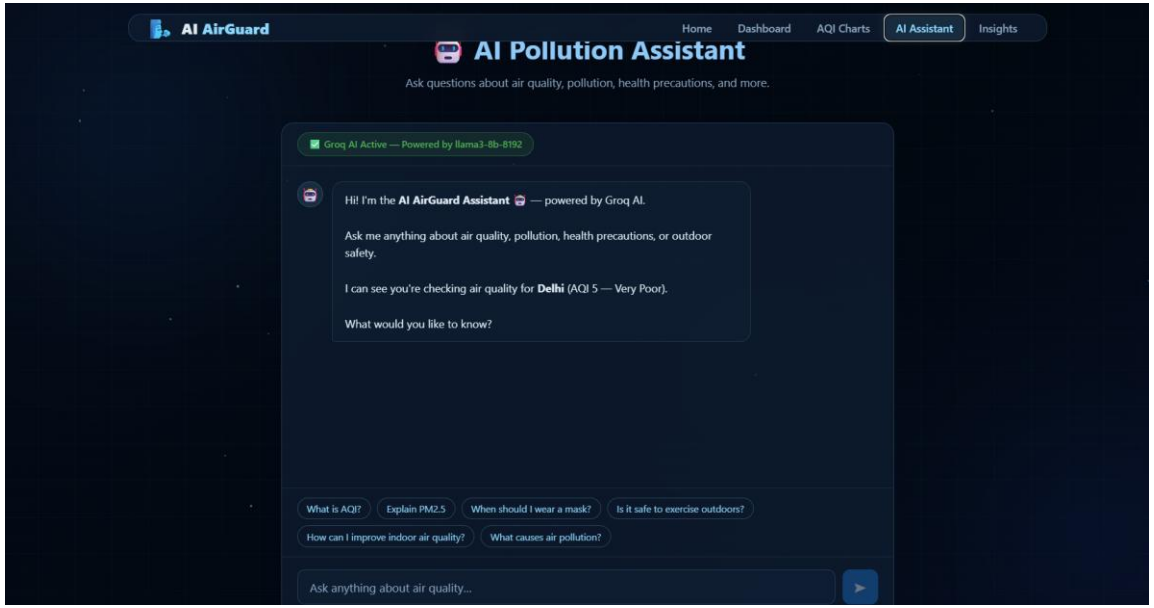




Project Output Screenshot







Novelty and Uniqueness

AI AirGuard goes beyond displaying air quality measurements by transforming pollution data into personalized, actionable recommendations. Its unique approach combines city-specific AQI monitoring, user-based health precautions, intelligent pollution alerts, and an interactive AI assistant in a single website. It helps users understand not only how polluted the air is, but also what actions they can take to reduce their exposure. The combination of air quality insights, personalized action plans, and easy-to-understand visualizations makes the solution practical, user-friendly, and valuable for everyday decision-making.

Git Hub Link

- **Github URL** –poojaarenad-6906/ai-airguard

(https://github.com/poojaarenad-6906/ai-airguard.git)

Future Scope

1. AI-Powered Air Quality Prediction –
The system can be enhanced with machine learning algorithms to predict future air quality levels using historical pollution data and weather conditions. This will help users take preventive measures before pollution reaches unhealthy levels.

2. AI-Powered Voice Assistant & Multilingual Support –
Enhancing the AI Assistant with voice interaction and support for multiple languages will improve accessibility. Users can ask questions about air pollution and receive health recommendations in their preferred language, making the system more user-friendly and inclusive.

3. Integration with Wearable & Health Devices –
The system can be integrated with smartwatches and fitness trackers to monitor activity levels and other available health metrics. This will enable more personalized recommendations based on air quality conditions and help users make safer decisions during outdoor activities.

Certificates Earned:

Please add your credly certificate





Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.